

CONSTRUCTION AND PROJECT MANAGEMENT I

Section B - Book Part: CPM & PERT

V. K. Kapoor textbook extracts mapped to term-final questions, 2016-2024

32

question-to-book links

479

linked examination marks

19 / 5 / 8

Exact / Direct / Partial

9 years

2016 through 2024

Key finding: The book extract is more complete than the lecture slides for CPM/PERT theory and worked examples. It directly supports 19 questions and 280 marks, with another 5 questions and 99 marks supported by the full transferable method.

Repeated textbook item	Question-bank years	Printed pages	Why it matters
Example 6: A-N deterministic CPM	2016	16.26-16.27	The activity table, four tasks, complete schedule and duration-change result are reproduced exactly.
Example 10: A-H probabilistic PERT	2017, 2019, 2024	16.49-16.50	Repeated three times with the same predecessors, estimates and 30-week / 99% requirement.
Example 11: A-L probabilistic PERT	2016, 2018, 2021, 2022	16.50-16.53	Repeated four times with total-float, critical-path and 99% completion requirements.

Classification: **Exact** = the requested definition, dataset or theory list is present; **Direct** = the complete solution method is present but the dataset/representation differs; **Partial** = only part of the answer is in the uploaded extracts.

1. Source Consolidation and Book Coverage

The three uploaded scans overlap. To avoid duplicate counting, the 16-page combined file is used as the master reference; the standalone CPM and PERT files are used to verify scans and supply the PERT opening on printed pages 16.38-16.39.

Uploaded source	Scan-page use	Printed book pages	Role in this report
CPM and PERT Book VK Kapoor.pdf	Master scan: 1-8 and 10-16	16.16-16.31 and 16.40-16.53	Primary page numbering, theory, examples and solutions.
CPM Book_VK Kapoor.pdf	10 scan pages	CPM extract, mainly 16.16-16.31	Cross-check for CPM definitions, floats, critical path and examples.
PERT Book_VK Kapoor.pdf	8 scan pages	16.38-16.53	Supplies the missing 16.38-16.39 PERT opening and verifies later PERT pages.

Coverage by printed page

Printed pages	Core content	Main exam use	Priority
16.16-16.17	Network objectives, notation, forward pass, ES and EF.	CPM definitions and the start of every numerical solution.	Highest
16.18-16.23	Backward pass; LS/LF; total, critical, free and independent float; worked calculation.	Seven years of float/slack/critical-path theory plus all CPM calculations.	Highest
16.24-16.31	CPM advantages; critical path method; worked Examples 5-8; Example 6 A-N.	2016 exact CPM problem and transferable numerical practice.	Highest
16.38-16.43	PERT introduction, three estimates, expected time, variance, probability and algorithm.	Definitions, formula questions and probability workflow.	Highest
16.44-16.45	Applications, advantages, shortcomings and distinction between PERT and CPM.	Exact theory answers in 2016-2018.	High
16.46-16.53	Normal table, Examples 9-11, A-H and A-L complete solutions.	Eight years of PERT numerical questions.	Highest

Not covered in the supplied book pages: GERT, LSM, Gantt-chart preparation, WBS/OBS, and full AON conventions. These components remain Partial rather than being filled from outside sources.

2.1 Detailed Question-to-Book Mapping: 2016-2018

Year / Q	Marks	QB page	Condensed question	Printed book page(s)	Match	Coverage assessment
2016 A 2(a)	4	1	Define slack time and critical path.	16.17-16.23	Exact	Definitions, zero-float condition and critical-path meaning are explicit.
2016 A 2(b)	6	1	Write the advantages and limitations of CPM.	16.24-16.25	Exact	The book gives a dedicated advantages section and practical limitations.
2016 A 2(c)	25	1	AOA CPM: paths, critical path, completion time, ES/EF/LS/LF/float, and changed duration.	16.26-16.27	Exact	Example 6 reproduces the A-N table, tasks, full schedule and changed-G result.
2016 A 3(a)	10	2	Distinguish between CPM and PERT.	16.44-16.45	Exact	A dedicated distinction list compares deterministic CPM with probabilistic PERT.
2016 A 3(b)	25	2	Twelve-activity PERT: floats, critical path and 99% completion time.	16.50-16.53	Exact	Example 11 reproduces the A-L dataset and the required 99% analysis.
2016 B 8(a)	15	3	Define PERT and GERT; state Gantt-chart objectives and preparation method.	16.38-16.45	Partial	PERT is covered fully; GERT and Gantt-chart preparation are absent from these scans.
2017 A 2(a)	15	4	Define total, free and independent float, critical path, early finish and late start.	16.17-16.23	Exact	All terms and formulas are developed together with a worked network.
2017 A 2(b)	20	4	Deterministic CPM network, dates/floats, critical path and duration change.	16.16-16.31	Direct	The complete method is worked through several examples; the exam dataset differs.
2017 A 4(a)	8	5	Define optimistic, most likely and pessimistic times and variance.	16.40-16.42	Exact	Three-time estimates, beta assumption, expected time and variance are explicit.
2017 A 4(b)	7	5	Write the advantages and shortcomings of PERT.	16.44-16.45	Exact	Both advantages and shortcomings are presented as separate lists.
2017 A 4(c)	20	5	A-H PERT: network, critical path and 99% confidence before 30 weeks.	16.49-16.50	Exact	Example 10 is the same A-H table and includes the probability/start-time solution.
2018 B 8(a)	10	8	Short notes on CPM, free float, variance, optimistic time and pessimistic time.	16.16-16.23; 16.40-16.42	Exact	Each requested item has a definition, formula or focused discussion.
2018 B 8(b)	5	8	Write the advantages and shortcomings of PERT.	16.44-16.45	Exact	Book headings directly match the requested theory answer.
2018 B 8(c)	20	8	A-L PERT: total float, critical path and 99% completion time.	16.50-16.53	Exact	Example 11 is the repeated A-L dataset and asks the same outputs.

QB page = page number in the uploaded 25-page question-bank PDF. Book page numbers are the printed V. K. Kapoor page numbers, not the scan-page numbers.

2.2 Detailed Question-to-Book Mapping: 2019-2021

Year / Q	Marks	QB page	Condensed question	Printed book page(s)	Match	Coverage assessment
2019 A 3(c)	15	9	Infer activity relationships and draw a network for a steelwork project.	16.16; 16.23-16.31	Partial	The book supplies arrow-network examples, but not the project-planning inference method.
2019 A 4(a)	10	9	Define free float, independent float, dummy activity, optimistic and pessimistic time.	16.20-16.21; 16.40-16.42	Partial	Float and PERT-time terms are exact; a standalone dummy-activity definition is not in the supplied pages.
2019 A 4(b)	25	10	A-H PERT: network, schedule, all floats and 99% start time.	16.49-16.50	Exact	Example 10 provides the same table, schedule and confidence calculation.
2020 A 2(a)	10	12	Define critical float, slack time, PERT, free float and independent float.	16.18-16.23; 16.38-16.42	Exact	The book covers zero-float criticality, slack/float types and PERT directly.
2020 A 2(b)	20	12	Draw network and calculate critical path, total, free and independent floats.	16.16-16.31	Direct	The full forward/backward-pass and float workflow is available with other datasets.
2020 A 3(c)	12	13	Draw an AOA network containing dummy activities.	16.16; 16.23-16.31	Partial	AOA examples use dummy links, but formal construction rules are not fully stated.
2021 A 3(c)	12	15	Draw the AOA network and mark the critical path.	16.16-16.31	Direct	Multiple AOA examples demonstrate network construction and critical-path identification.
2021 A 4(c)	20	16	A-L PERT: total float, critical path and 99% completion time.	16.50-16.53	Exact	Exact textbook Example 11 repeat.
2021 B 8(b)	8	17	Define GERT and LSM; differentiate GERT from CPM or PERT.	16.38-16.45	Partial	CPM and PERT are covered; GERT and LSM are outside the uploaded book pages.

QB page = page number in the uploaded 25-page question-bank PDF. Book page numbers are the printed V. K. Kapoor page numbers, not the scan-page numbers.

2.3 Detailed Question-to-Book Mapping: 2022-2024

Year / Q	Marks	QB page	Condensed question	Printed book page(s)	Match	Coverage assessment
2022 A 4(a)	25	18	A-L PERT: floats, critical path and 99% completion time.	16.50-16.53	Exact	Exact textbook Example 11 repeat.
2022 B 8(b)	8	19	Define GERT and differentiate GERT from PERT.	16.38-16.45	Partial	PERT is supported, but GERT is not present.
2023 A 2(b)	12	20	Short notes on WBS, OBS, AOA network and project planning.	16.16-16.31	Partial	Only the AOA/network portion belongs to the supplied book extract.
2023 A 3(b)	22	20	AON CPM: paths, critical path, duration, ES/EF/LS/LF/float and changed duration.	16.16-16.31	Direct	All calculations are transferable, but the book examples are AOA rather than AON.
2023 A 4(a)	25	20	Eleven-activity PERT: floats, critical path and 99% completion time.	16.38-16.53	Direct	The complete method and two close worked problems are supplied; data differ.
2023 A 4(b)	10	21	Define total, free and independent float, early start and late start.	16.17-16.23	Exact	Definitions and equations are explicit.
2024 A 3(b)	20	23	Draw AOA and AON networks from predecessor information.	16.16-16.31	Partial	AOA is supported; AON conventions are not covered in the supplied pages.
2024 A 4(a)	10	23	Define critical path, free float, independent float, variance and slack time.	16.17-16.23; 16.40-16.42	Exact	All requested concepts and formulas are present.
2024 A 4(b)	25	24	A-H PERT: network, critical path, schedule and 99% start time.	16.49-16.50	Exact	Exact textbook Example 10 repeat.

QB page = page number in the uploaded 25-page question-bank PDF. Book page numbers are the printed V. K. Kapoor page numbers, not the scan-page numbers.

3.1 Exact Textbook Repeat: 2016 Deterministic CPM

Question 2016 A2(c) is taken directly from Example 6. The A-N predecessors and durations, four requested tasks, calculated schedule and changed duration of activity G are all present on printed pages 16.26-16.27.

16-26

CONTEMPORARY OPERATIONS RESEARCH

It may be noted from the above diagram that critical path comprises the activities (1,2), (2,3), (3,5) and (5,6) and this path represents the minimum possible time to complete the entire project.

COMPUTATION OF FLOATS

Activity	Duration	Earliest		Latest		Float	
i-j		Start	Finish	Start	Finish	Total	Free
A 1-2	8	0	8	0	8	0	0
B 1-3	10	0	10	8	18	8	8
C 1-4	8	0	8	9	17	9	0
D 2-3	10	8	18	8	18	0	0
E 2-6	16	8	24	28	44	20	20
F 3-5	17	18	35	18	35	0	0
G 4-5	18	8	26	17	35	9	9
H 4-6	14	8	22	30	44	22	22
I 5-6	9	35	44	35	44	0	0

Example 6. A project has the following characteristics :

Activity	Preceding activity	Expected completion time (in weeks)
A	None	5
B	A	2
C	A	6
D	B	12
E	D	10
F	D	9
G	D	5
H	B	9
I	C, E	1
J	G	2
K	F, I, J	3
L	K	9
M	H, G	7
N	M	8

(i) Draw a PERT network for this project.
(ii) Find the various paths and the critical path as well as the project completion time.
(iii) Prepare an activity schedule showing the ES, EF, LS, LF, and float for each activity.
(iv) Will the critical path change if activity G takes 10 weeks instead of 5 weeks? If so, what will be the new critical path?

(Delhi Univ. M. Com., 1987)

16-27

PERT AND CPM-I

Solution. The network for the given project is shown in Figure 16-7

The various paths and their lengths are as follows :

Path	Duration (Weeks)
I. 1-2-5-8-10-11	24
II. 1-2-3-7-9-11	31
III. 1-2-3-4-6-8-10-11	38
IV. 1-2-3-4-6-7-9-11	39
V. 1-2-3-4-8-10-11	40
VI. 1-2-3-4-5-8-10-11	42

Thus, the critical path is 1-2-3-4-5-8-10-11 and the duration of the project is 42 weeks.

(iii) DETERMINATION OF EARLIEST AND LATEST TIMES

Activity	Duration	ES	EF	LS	LF	Float
A	5	0	5	0	5	0
B	2	5	7	5	7	0
C	6	5	11	5	11	0
D	12	7	19	7	19	0
E	10	19	29	19	29	0
F	9	19	28	19	29	0
G	5	19	24	21	30	2
H	9	7	16	18	27	3
I	1	29	30	29	30	0
J	2	24	26	28	30	0
K	3	30	33	30	33	0
L	9	33	42	33	42	0
M	7	24	31	27	34	3
N	8	31	39	34	42	3

(iv) If activity G (4-6) takes 10 weeks instead of 5, the lengths of the paths III and IV would become 43 and 44 weeks respectively. In that case path 1-2-3-4-6-7-9-11 would become critical.

V. K. Kapoor, printed pages 16.26-16.27: Example 6 and complete solution.

(c) Explain construction project organization process. (08)

(c) What do you mean by project organization? Discuss different types of project organization. (09)

(d) Write down the pros and cons of the following: (09)

- Functional organization
- Projectized organization
- Matrix organization

2. (a) Define: (i) Slack Time (ii) Critical Path (04)

(b) Write down the advantages and limitation of CPM technique. (06)

(c) A project has the following characteristics: (25)

Activity	Preceding activity	Expected completion time (weeks)
A	None	5
B	A	2
C	A	6
D	B	12
E	D	10
F	D	9
G	D	5
H	B	9
I	C, E	1
J	G	2
K	F, I, J	3
L	K	9
M	H, G	7
N	M	8

(i) Draw a AOA network for this project.
(ii) Find the various paths and the critical path as well as the project completion time.
(iii) Prepare an activity schedule showing the ES, EF, LS, LF and float.
(iv) Will the critical path change if activity G takes 10 weeks instead of 5 weeks? If so, what will be the new critical path?

Page 1 of 3

Question bank 2016, PDF page 1: Q2(a-c).

Verified textbook result: the original critical path is 1-2-3-4-5-8-10-11 and project duration is 42 weeks. If activity G (4-6) changes from 5 to 10 weeks, path 1-2-3-4-6-7-9-11 becomes critical with a 45-week duration.

4. Reverse Index, Formula Sheet and Exam Strategy

Book pages	Topic	Mapped questions
16.16-16.17	Network objectives, notation and forward pass	2016 A2(c); 2017 A2(b); 2020 A2(b); 2021 A3(c); 2023 A3(b); 2024 A3(b)
16.18-16.23	Backward pass, float/slack types and critical path	2016 A2(a-c); 2017 A2(a-b); 2018 B8(a); 2019 A4(a); 2020 A2(a-b); 2023 A4(b); 2024 A4(a)
16.24-16.31	CPM advantages and deterministic examples	2016 A2(b-c); 2017 A2(b); 2020 A2(b); 2021 A3(c); 2023 A3(b)
16.38-16.45	PERT foundation, algorithm, theory, pros/cons and CPM distinction	2016 A3(a), B8(a); 2017 A4(a-b); 2018 B8(a-b); 2020 A2(a); 2021/2022 GERT comparisons (partial); 2024 A4(a)
16.46-16.53	Normal distribution and Examples 9-11	2016 A3(b); 2017 A4(c); 2018 B8(c); 2019 A4(b); 2021 A4(c); 2022 A4(a); 2023 A4(a); 2024 A4(b)

Purpose	Book rule / formula	Exam use
Expected activity time	$te = (a + 4m + b) / 6$	Calculate one duration for each probabilistic activity.
Activity variance	$variance = [(b - a) / 6]^2$	Sum only critical-path variances for project uncertainty.
Forward pass	$EF = ES + duration$; $ES = \text{maximum predecessor EF}$	Earliest dates and project duration.
Backward pass	$LS = LF - duration$; $LF = \text{minimum successor LS}$	Latest dates without delaying completion.
Total float	$TF = LS - ES = LF - EF$	Zero identifies a critical activity.
Deadline probability	$Z = (\text{due date} - \text{expected duration}) / \text{project SD}$	Read the cumulative probability from the normal table.
99% duration	$\text{Expected duration} + 2.33 \times \text{project SD}$	Reverse the calculation to find how early work must start.

Recommended revision order: (1) pages 16.17-16.23 definitions and float formulas; (2) Example 6 pages 16.26-16.27; (3) PERT formulas pages 16.40-16.43; (4) theory pages 16.44-16.45; (5) Examples 10 and 11 pages 16.49-16.53. Keep separate notes for GERT, LSM, Gantt charts, WBS/OBS and AON conventions.